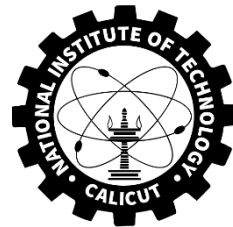


# **Quality, Reliability & Maintenance ME3327E**



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**Course Instructor:**

**Dr. Nehem Tudu**

**Assistant Professor, Dept. of Mechanical Engineering**

# Operating Characteristic Curve

- An operating characteristic (OC) curve is a measure of goodness of a control chart's ability to detect changes in process parameters.
- Specifically, it is a plot of the probability of the type II error versus the shift of a process parameter value from its in-control value.
- OC curves enable us to determine the chances of not detecting a shift of a certain magnitude in a process parameter on a control chart.

**Exercise 3:** Refer to the data in Exercise 2, involving the control chart for the average breaking strength of nylon fibers. Samples of size 5 are randomly chosen from a process whose mean and standard deviation are estimated to be 120 kg and 8 kg, respectively. Construct the operating characteristic curve for increases in the process mean from 120 kg.

# Exercise

A control chart is to be constructed for the average breaking strength of nylon fibers. Samples of size 5 are randomly chosen from the process. The process mean and standard deviation are estimated to be 120 kg and 8 kg, respectively.

- If the control limits are placed three standard deviations from the process mean, what is the probability of a type I error?
- If the process mean shifts to 125 kg, what is the probability of concluding that the process is in control and hence making a type II error on the first sample plotted after the shift?
- What is the probability of detecting the shift by the second sample plotted after the shift if the samples are chosen independently?

Refer to the data in Exercise 2, involving the control chart for the average breaking strength of nylon fibers. Samples of size 5 are randomly chosen from a process whose mean and standard deviation are estimated to be 120 kg and 8 kg, respectively. Construct the operating characteristic curve for increases in the process mean from 120 kg.

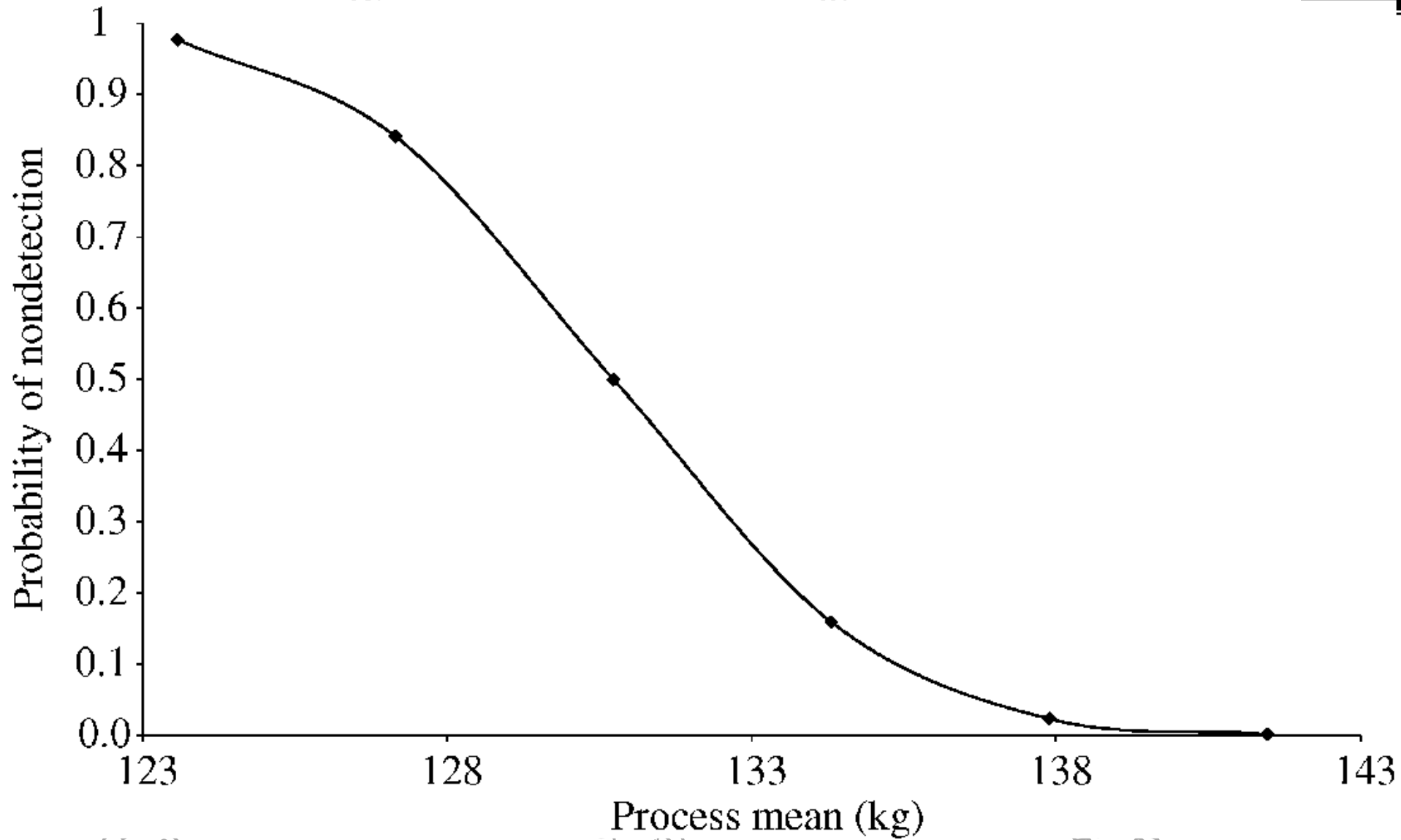
# Solution



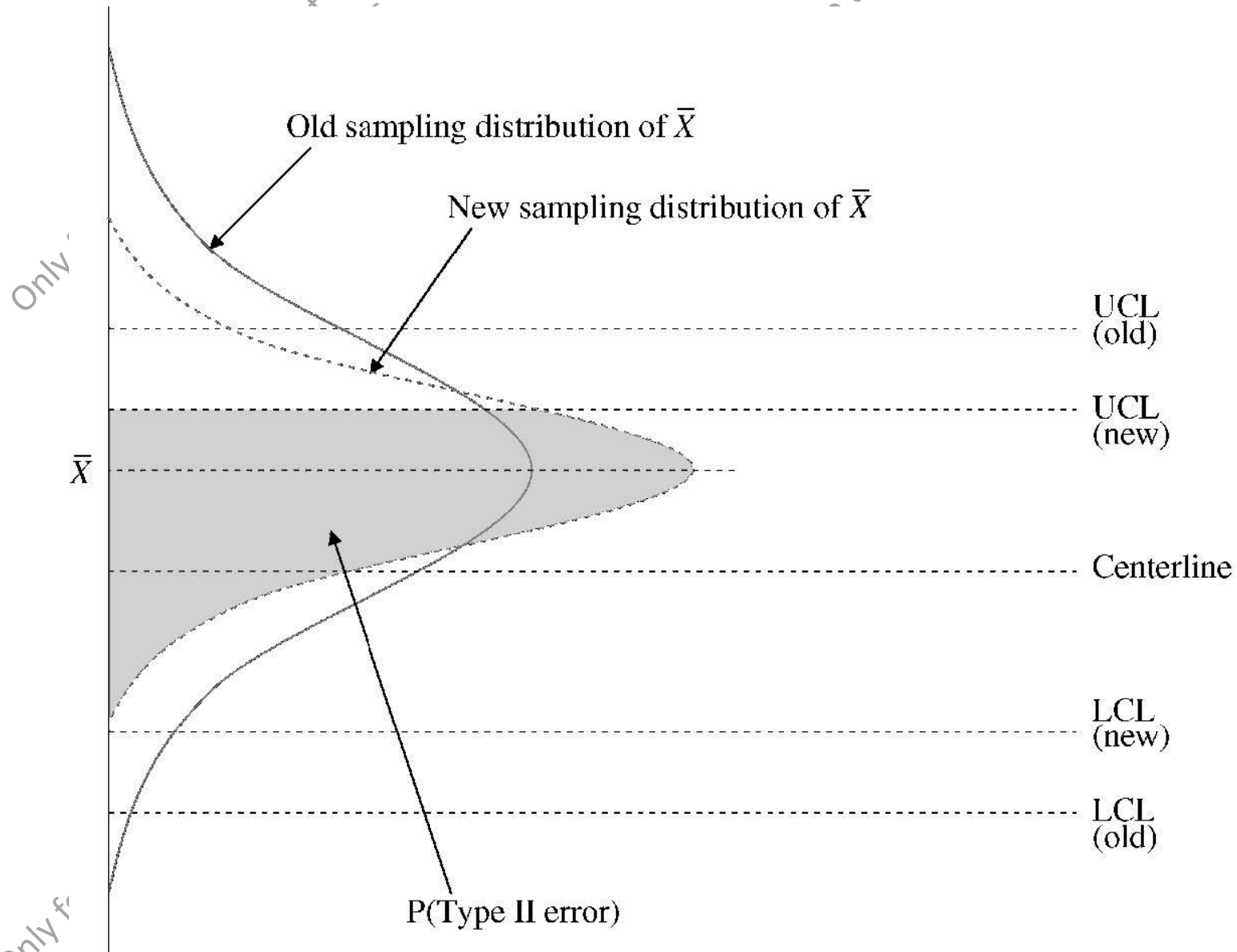
| Process Mean | Z-Value<br>at UCL, $Z_1$ | Area Above<br>UCL | Z-Value<br>at LCL, $Z_2$ | Area Below<br>LCL | Probability of<br>Nondetection, $\beta$ |
|--------------|--------------------------|-------------------|--------------------------|-------------------|-----------------------------------------|
| 123.578      | 2.00                     | 0.0228            | -4.00                    | 0.0000            | 0.9772                                  |
| 127.156      | 1.00                     | 0.1587            | -5.00                    | 0.0000            | 0.8413                                  |
| 130.733      | 0.00                     | 0.5000            | -6.00                    | 0.0000            | 0.5000                                  |
| 134.311      | -1.00                    | 0.8413            | -7.00                    | 0.0000            | 0.1587                                  |
| 137.888      | -2.00                    | 0.9772            | -8.00                    | 0.0000            | 0.0228                                  |
| 141.466      | -3.00                    | 0.9987            | -9.00                    | 0.0000            | 0.0013                                  |



# Solution



# Effect of Control Limits on Errors in Inference Making



- **Warning limits** are usually placed at two standard deviations from the center line.
- **Effect of Sample Size on Control Limits**
  - The sample size usually has an influence on the standard deviation of the sample statistic being plotted on the control chart.

- **Average Run Length**

- An alternative measure of the performance of a control chart, in addition to the OC curve, is the average run length (ARL). This denotes the number of samples, on average, required to detect an out-of-control signal.

Let  $P_d$  denote the probability of an observation plotting outside the control limits.

$$ARL = \frac{1}{P_d}$$

# Selection of Rational Samples

Shewhart described the fundamental criteria for the selection of rational subgroups, or rational samples. A rational sample is chosen in such a manner that the variation within it is considered to be due only to common causes. So, samples are selected such that if special causes are present, they will occur between the samples. Therefore, the differences between samples will be maximized, and differences within samples will be minimized.

**Sample Size:** Selecting sample size the number of items in each sample is a necessity in using control charts. The degree of shift in the process parameter expected to take place will influence the choice of sample size.

**Frequency of Sampling:** The sampling frequency must be decided prior to the construction of control charts. Choosing large samples very frequently is the sampling scheme that provides the most information.

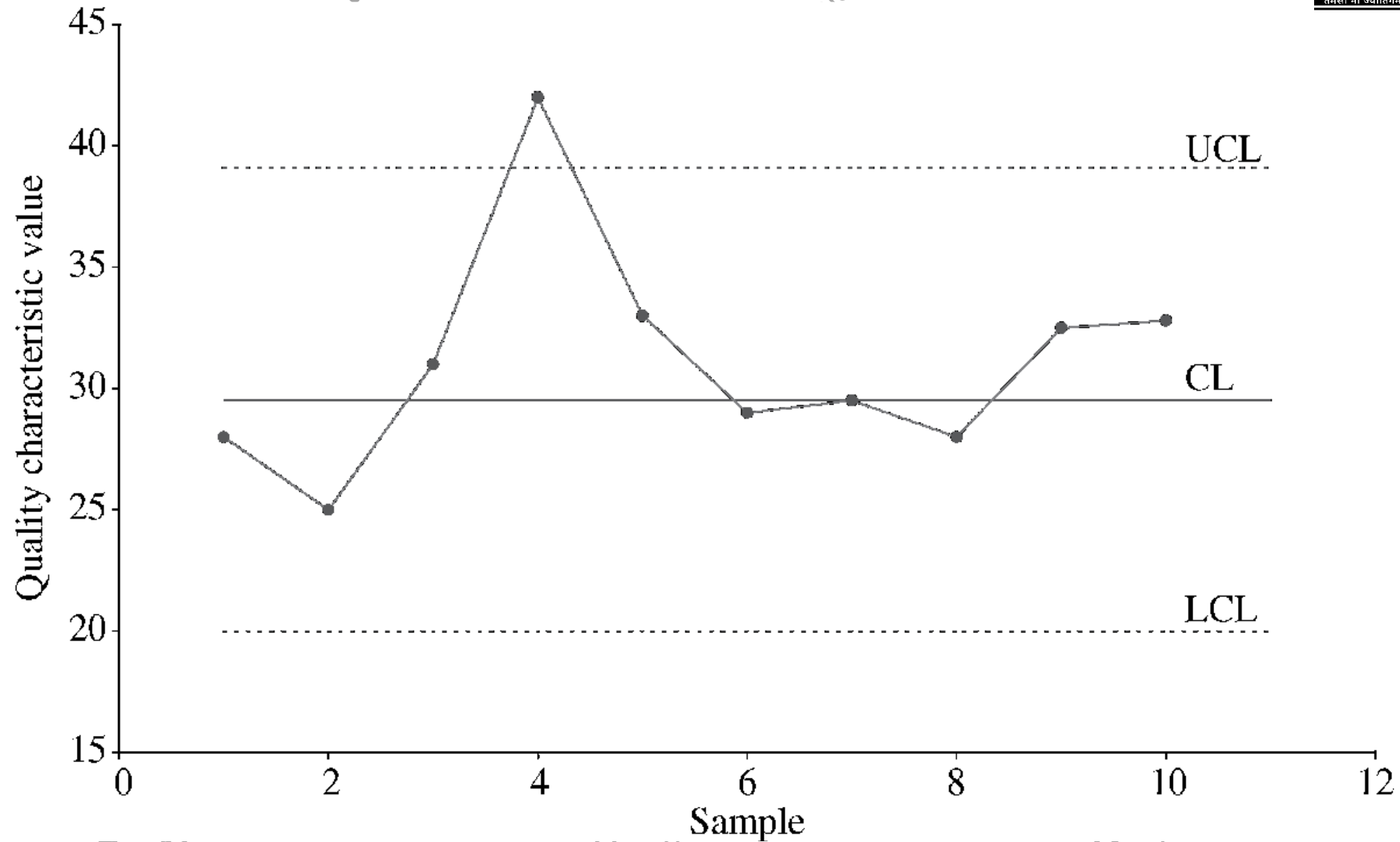
# Analysis of Patterns in Control Charts

One of the main objectives of using control charts is to determine when a process is out of control so that necessary actions may be taken.

## Some Rules for Identifying an Out-of-Control Process

**Rule 1:** A process is assumed to be out of control if a single point plots outside the control limits.

# Is it in-control or out-of-control



Only for  $S_{\bar{X}}$  in an,

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# Analysis of Patterns in Control Charts

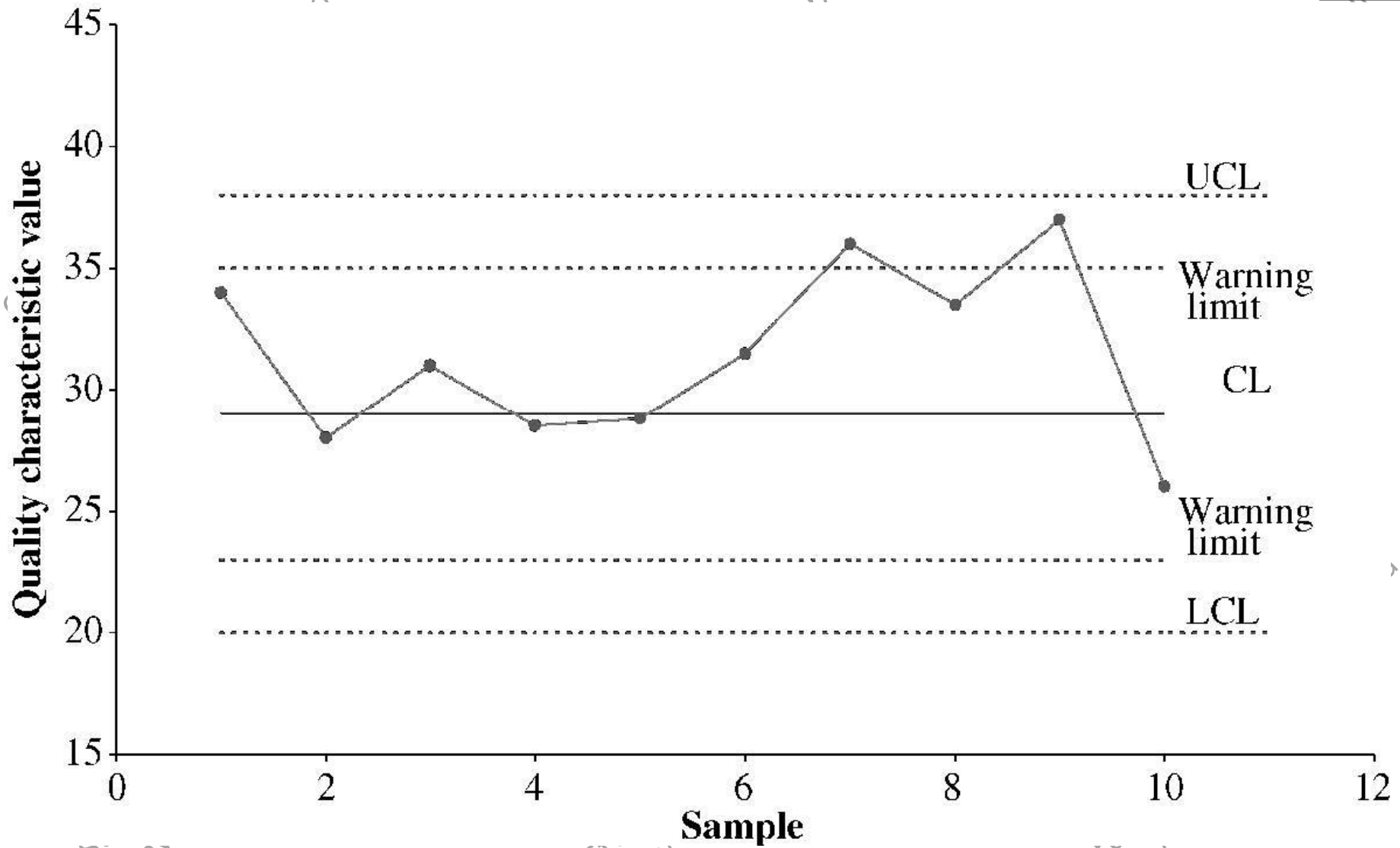
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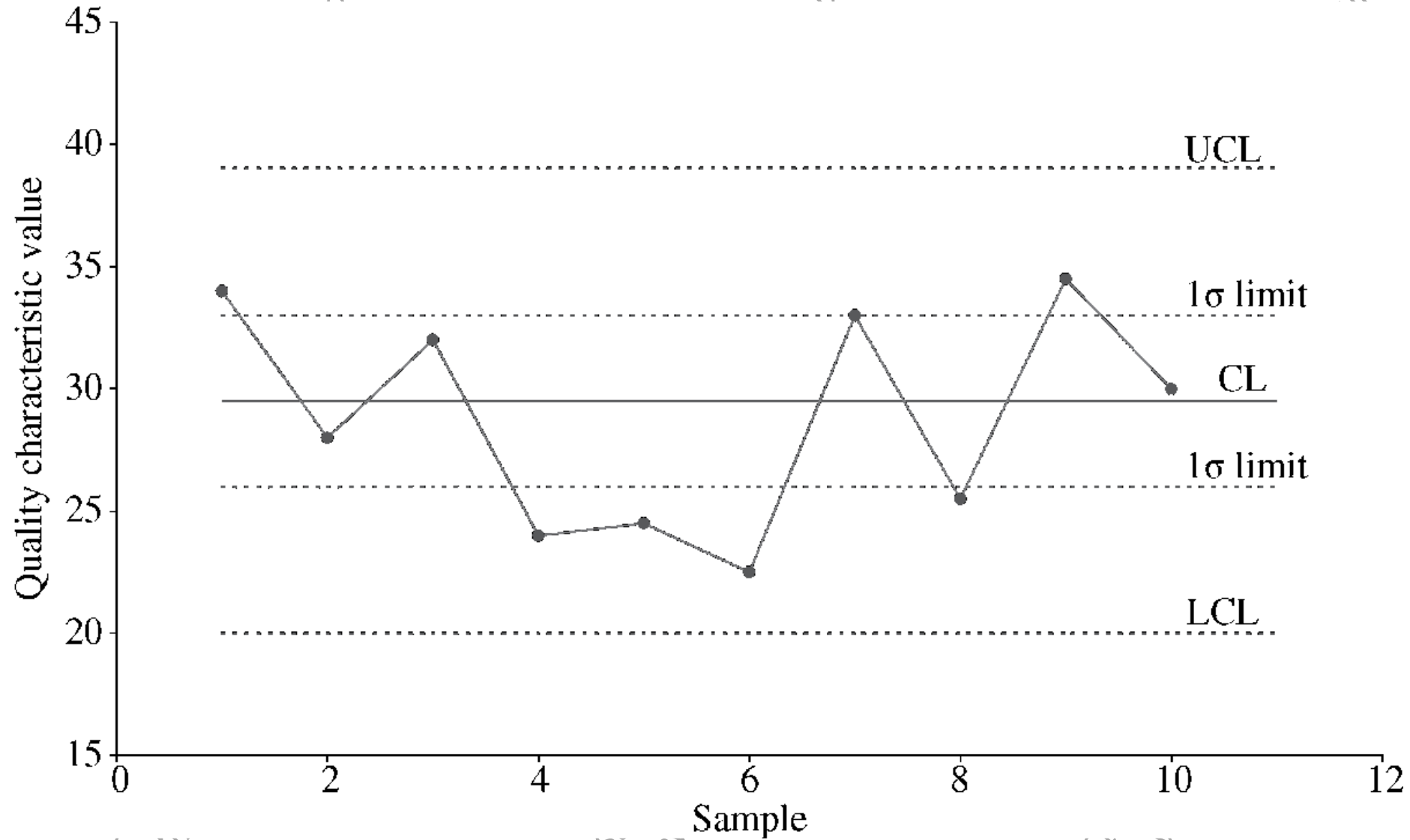
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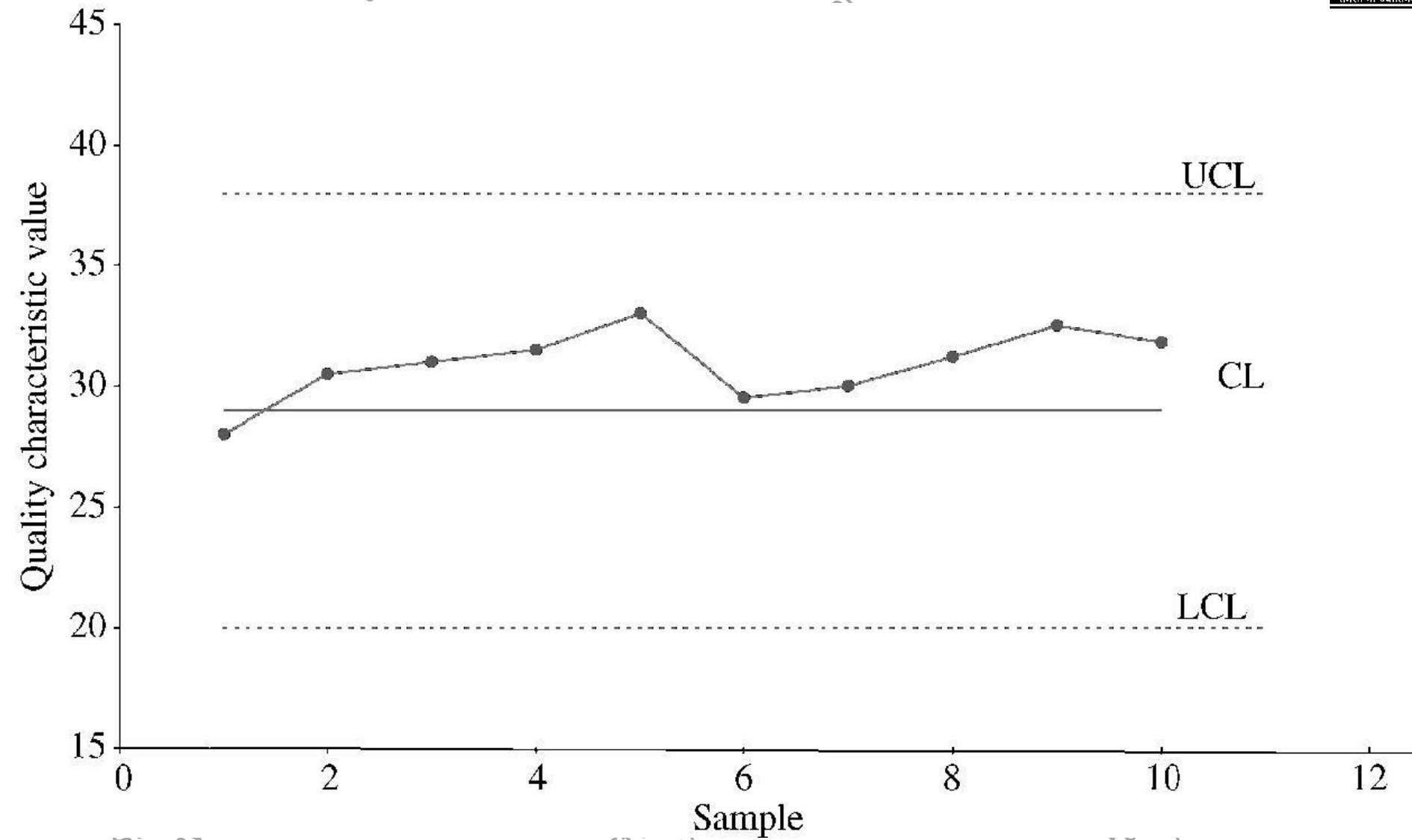
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**Rule 4:** A process is assumed to be out of control if nine or more consecutive points fall to one side of the center line.

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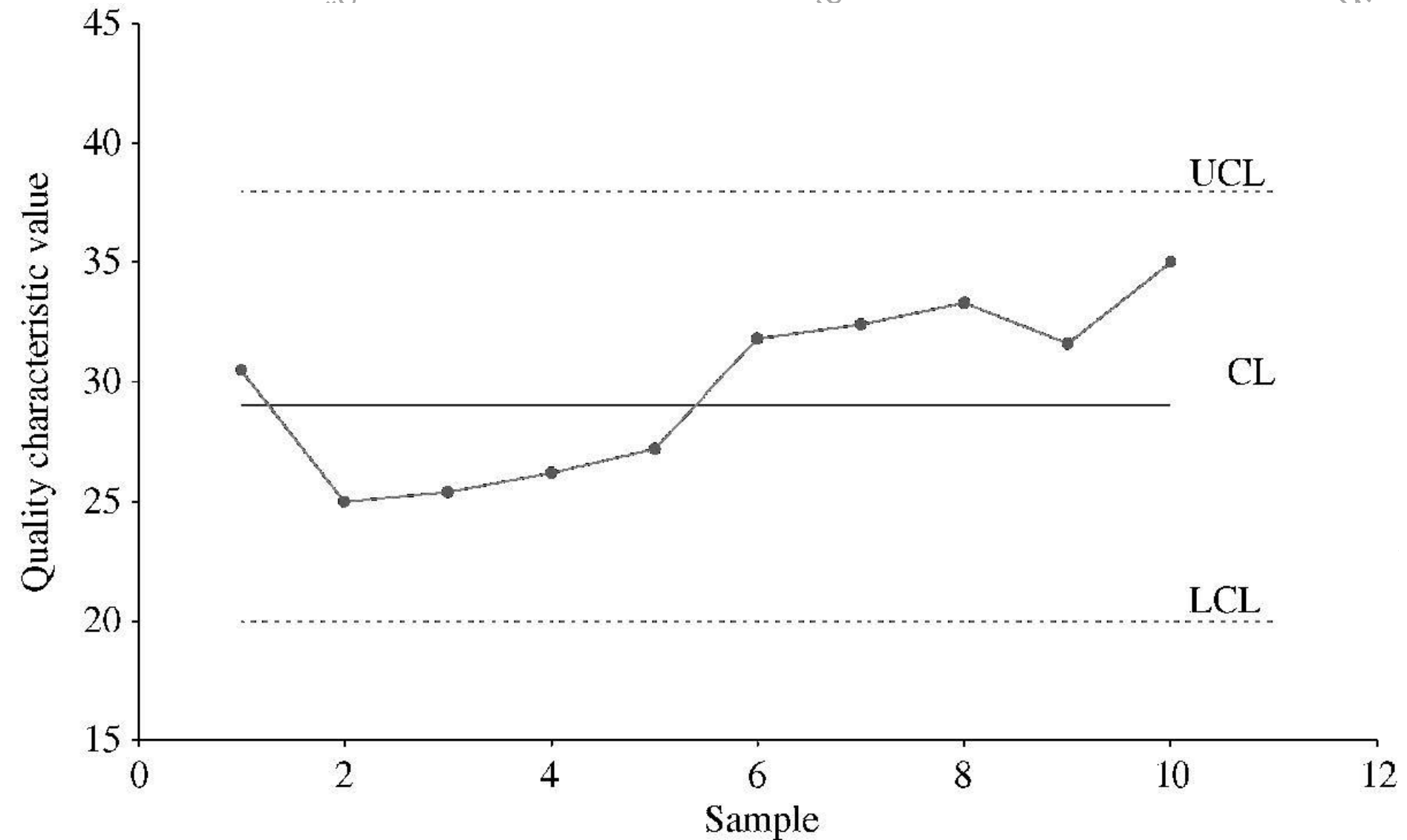
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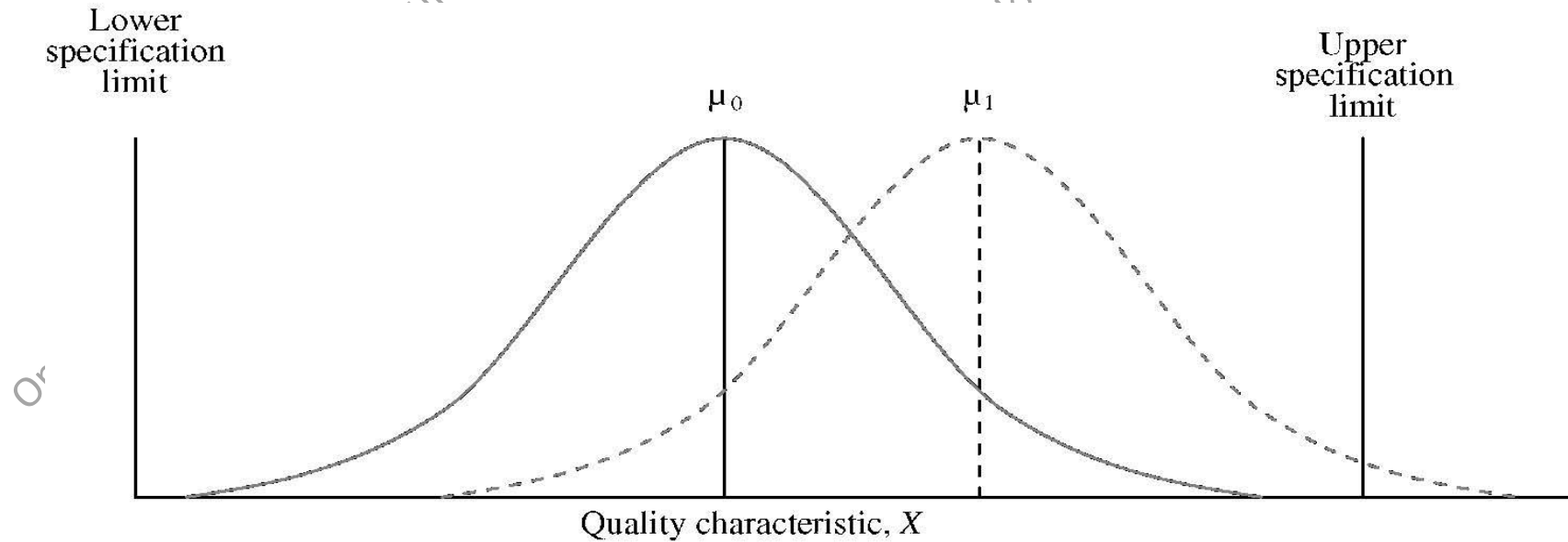
**Rule 4:** A process is assumed to be out of control if nine or more consecutive points fall to one side of the center line.

**Rule 5:** A process is assumed to be out of control if there is a run of six or more consecutive points steadily increasing or decreasing.

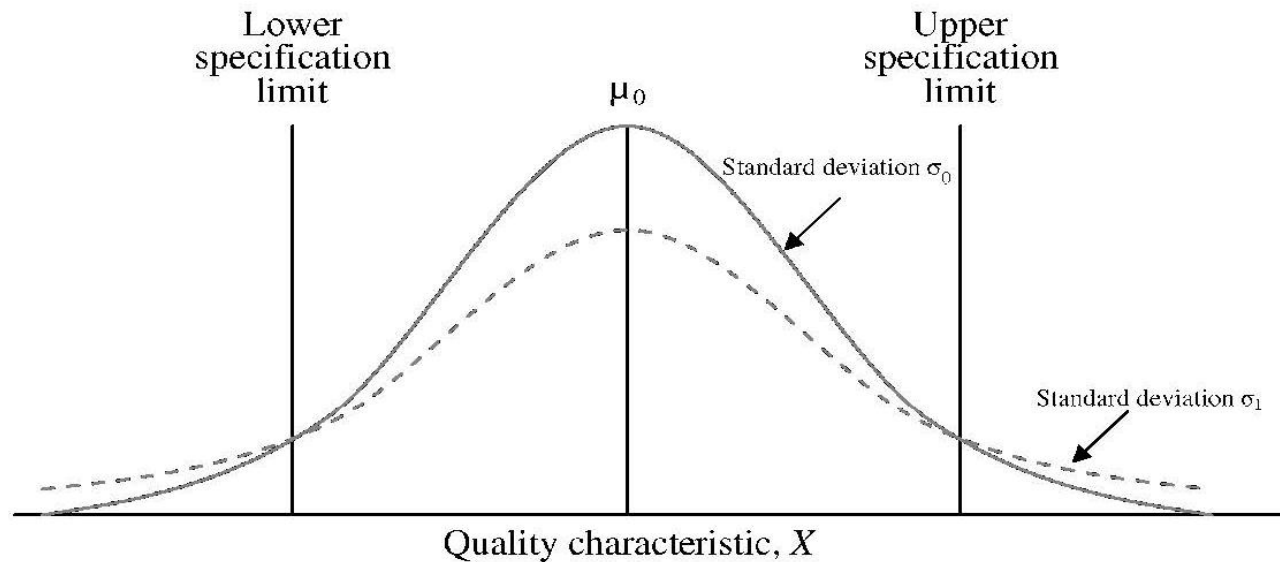
# Is it in-control or out-of-control



# Changes in the mean and dispersion of a process



(a) Change in the process mean of a quality characteristic



(b) Change in the dispersion of a quality characteristic

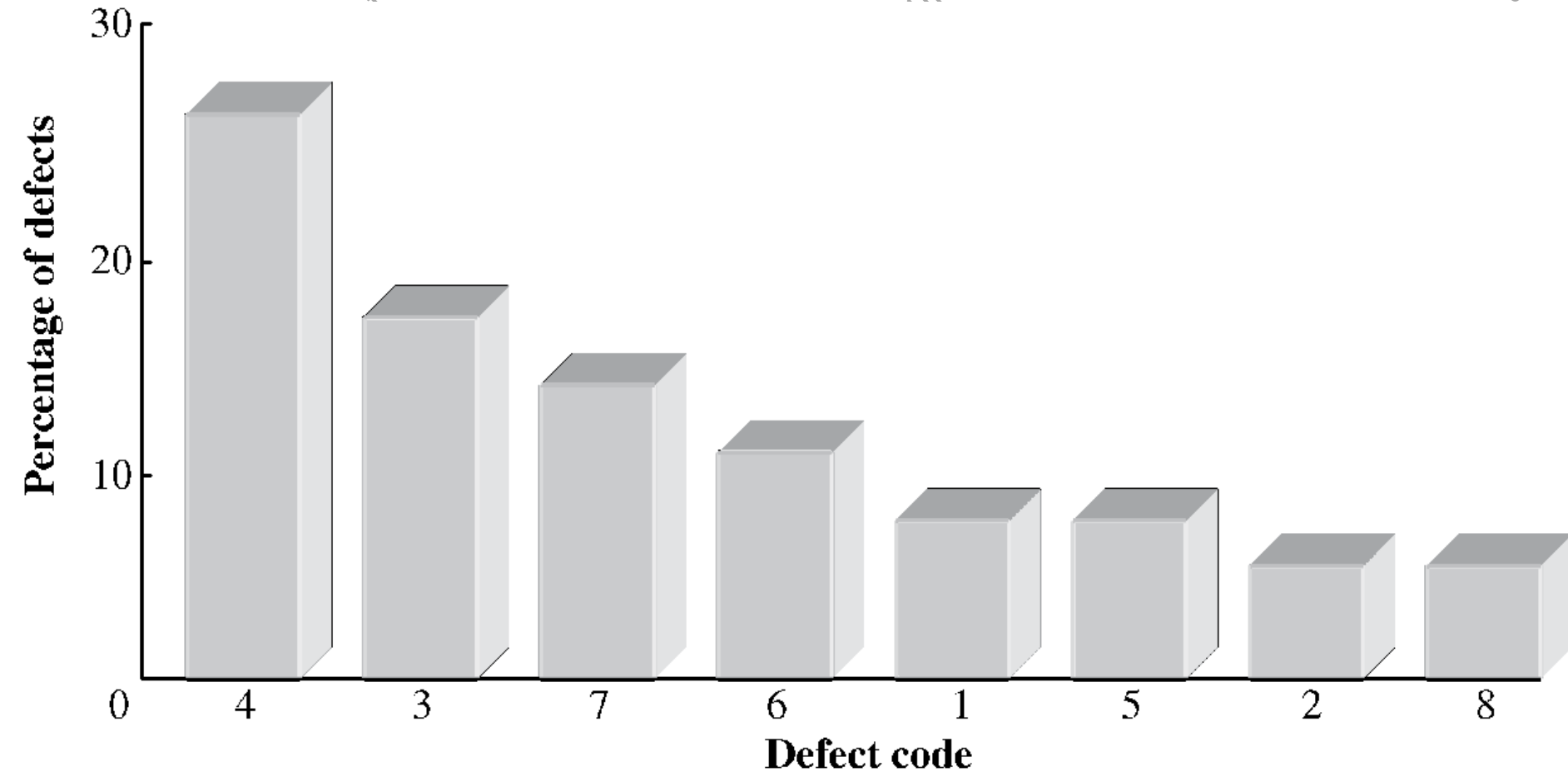
# Selection of Characteristics for Investigation

## Pareto Analysis of Defects for Assembly Data

| Defect Code | Defect                                         | Frequency | %     |
|-------------|------------------------------------------------|-----------|-------|
| 1           | Outside diameter of hub                        | 30        | 8.82  |
| 2           | Depth of key way                               | 20        | 5.88  |
| 3           | Hub length                                     | 60        | 17.65 |
| 4           | Inside diameter of hub                         | 90        | 26.47 |
| 5           | Width of key way                               | 30        | 8.82  |
| 6           | Thickness of flange                            | 40        | 11.77 |
| 7           | Depth of slot                                  | 50        | 14.71 |
| 8           | Hardness (measured by Brinell Hardness number) | 20        | 5.88  |



# Selection of Characteristics for Investigation



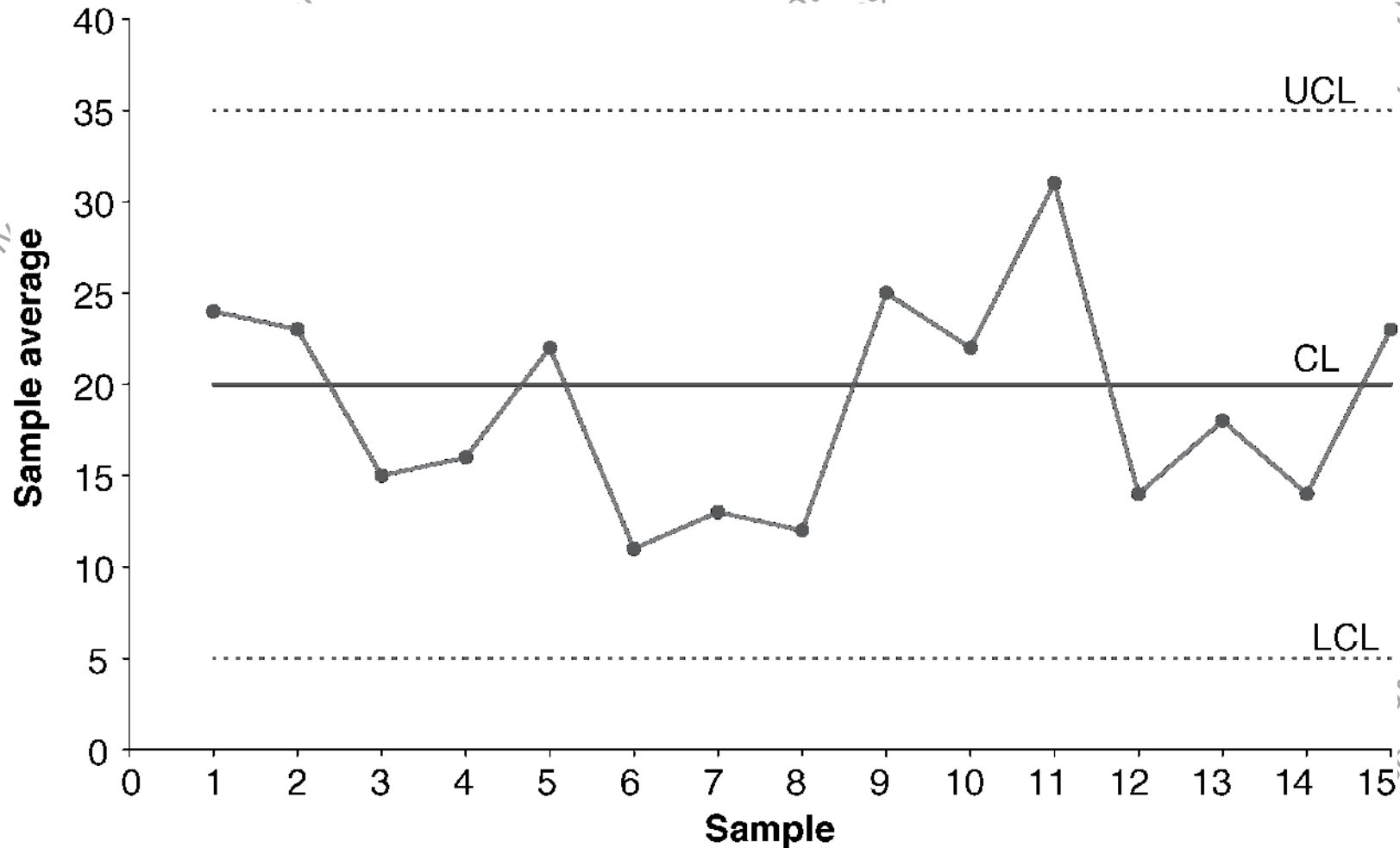
Pareto diagram for assembly data

# Analysis of Patterns in Control Charts

- Selection of Rational Samples
- Sample Size
- Frequency of Sampling
- Choice of Measuring Instruments
- Design of Data Recording Forms

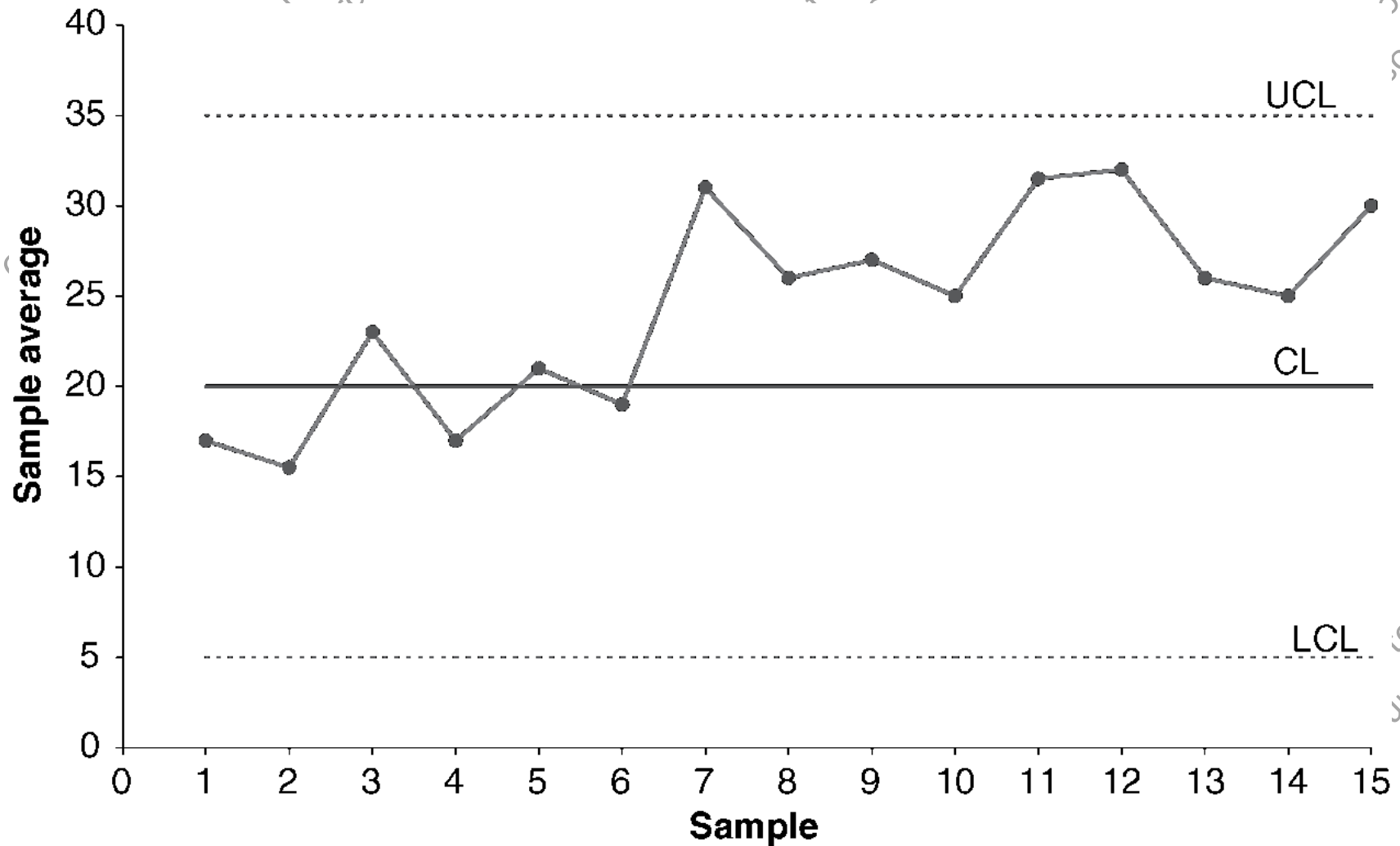
# Control Chart Patterns and Corrective

- Natural Patterns



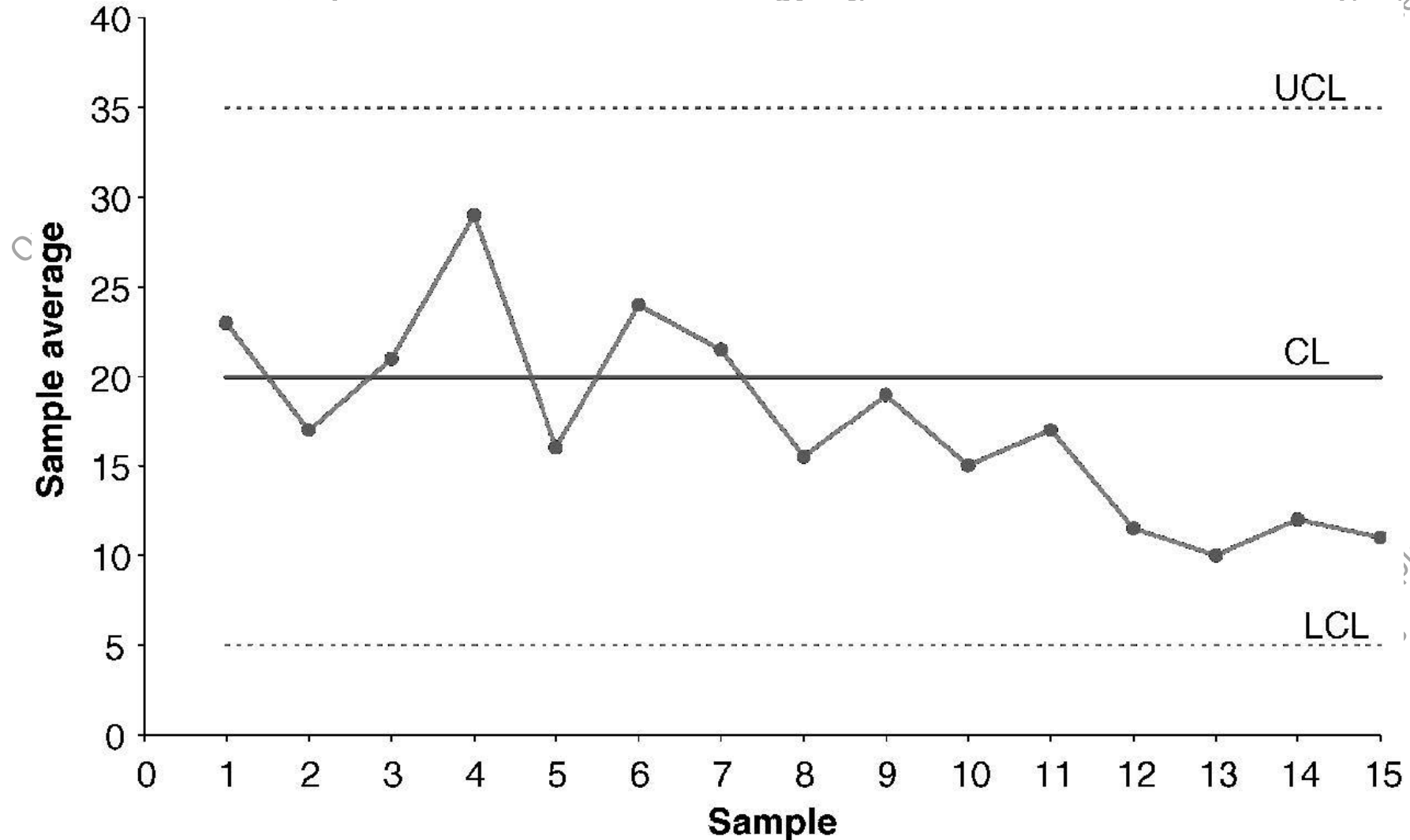
# Control Chart Patterns and Corrective

- Sudden Shifts in the Level



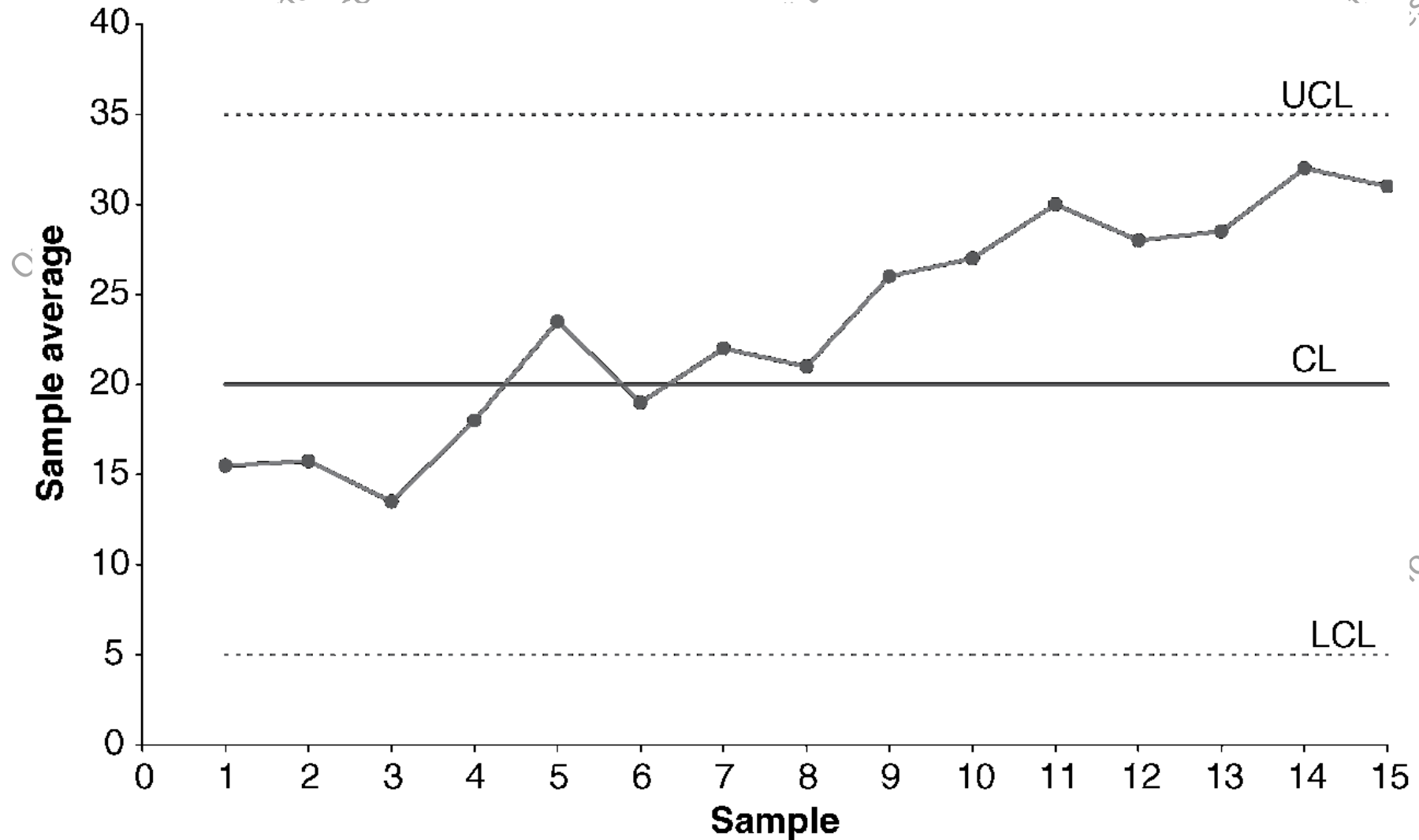
# Control Chart Patterns and Corrective

- Gradual Shifts in the Level



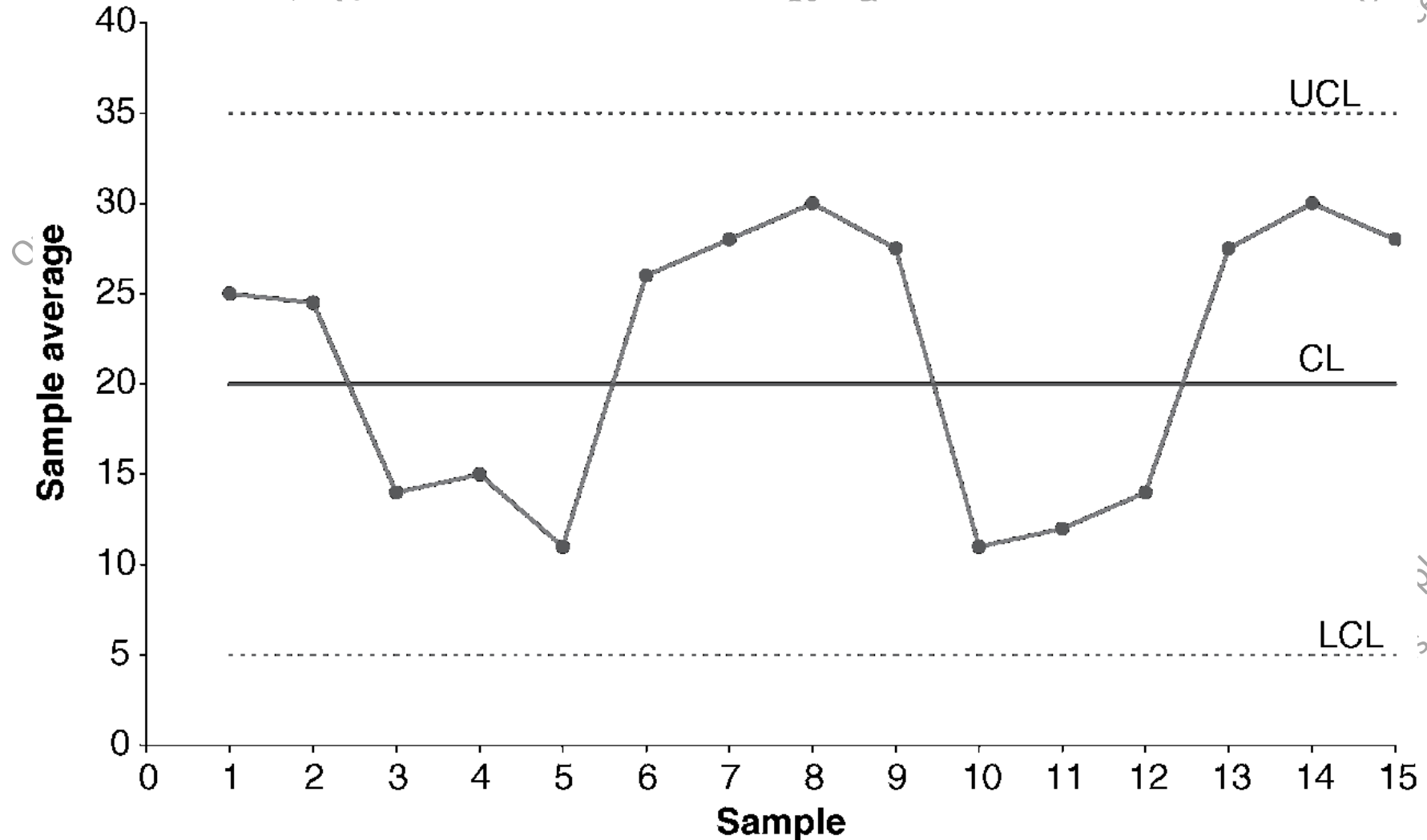
# Control Chart Patterns and Corrective

- Trending Pattern



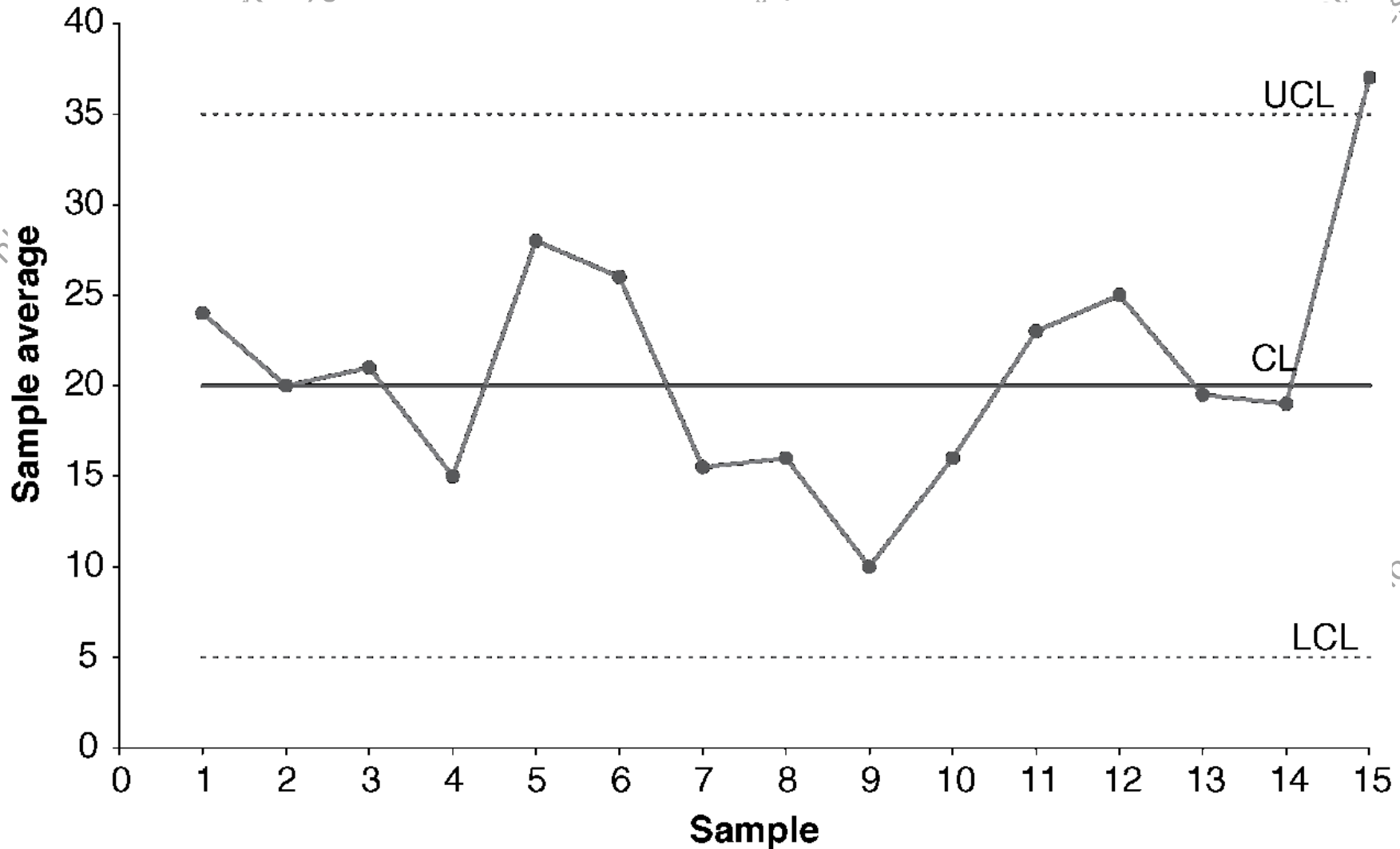
# Control Chart Patterns and Corrective

- Cyclic Patterns



# Control Chart Patterns and Corrective

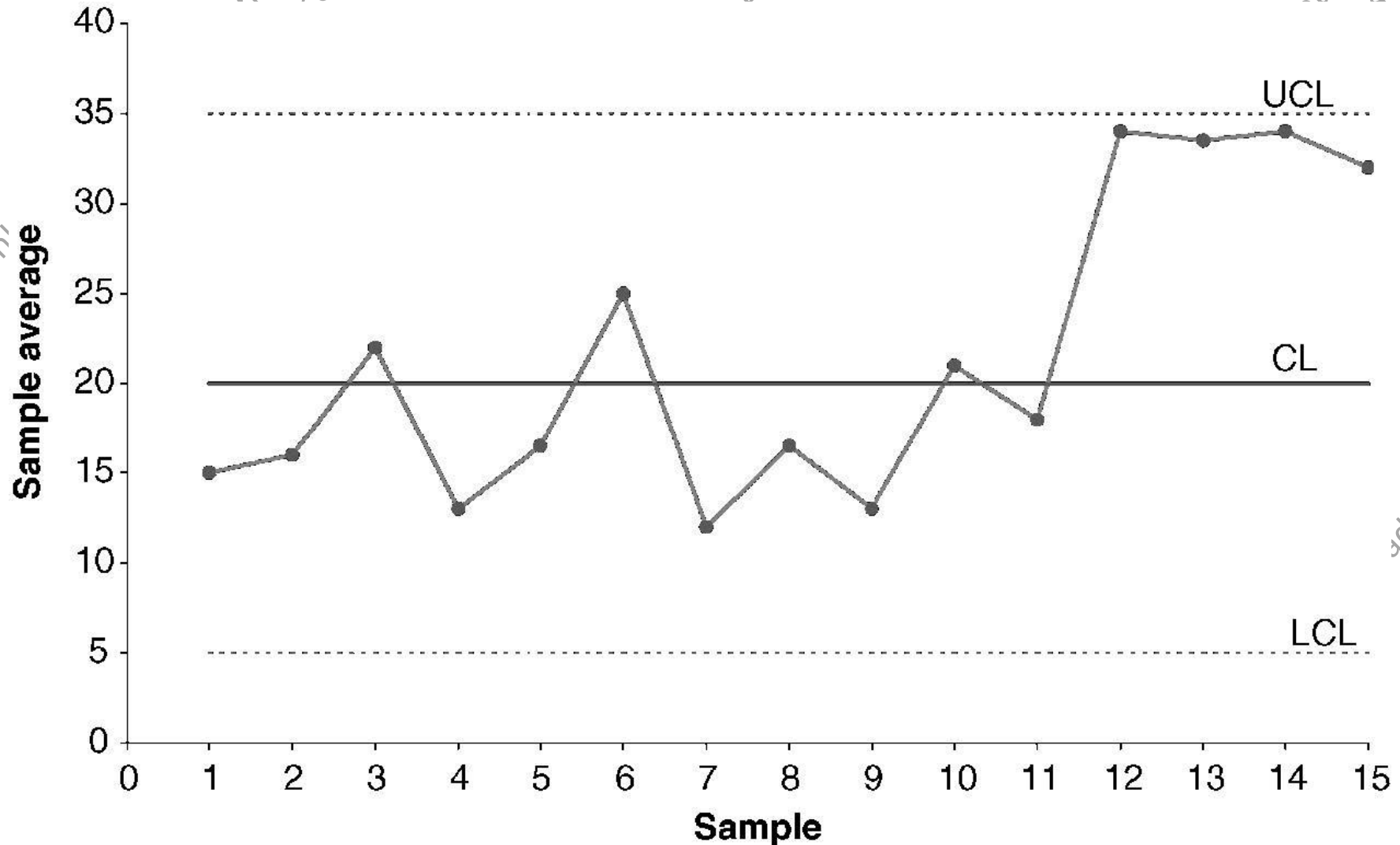
- Wild Patterns – Freaks Pattern





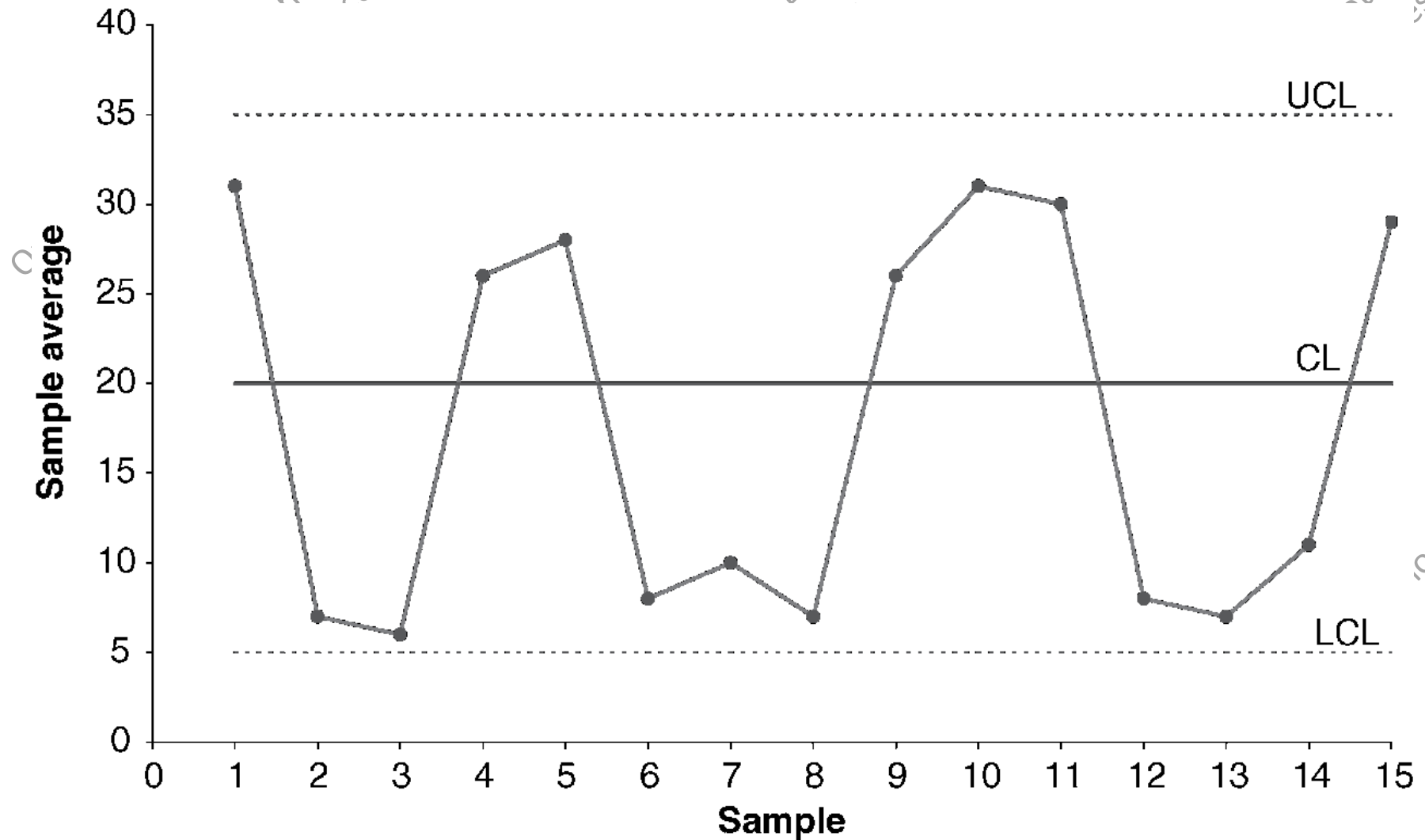
# Control Chart Patterns and Corrective

- Wild Patterns – Bunch Pattern



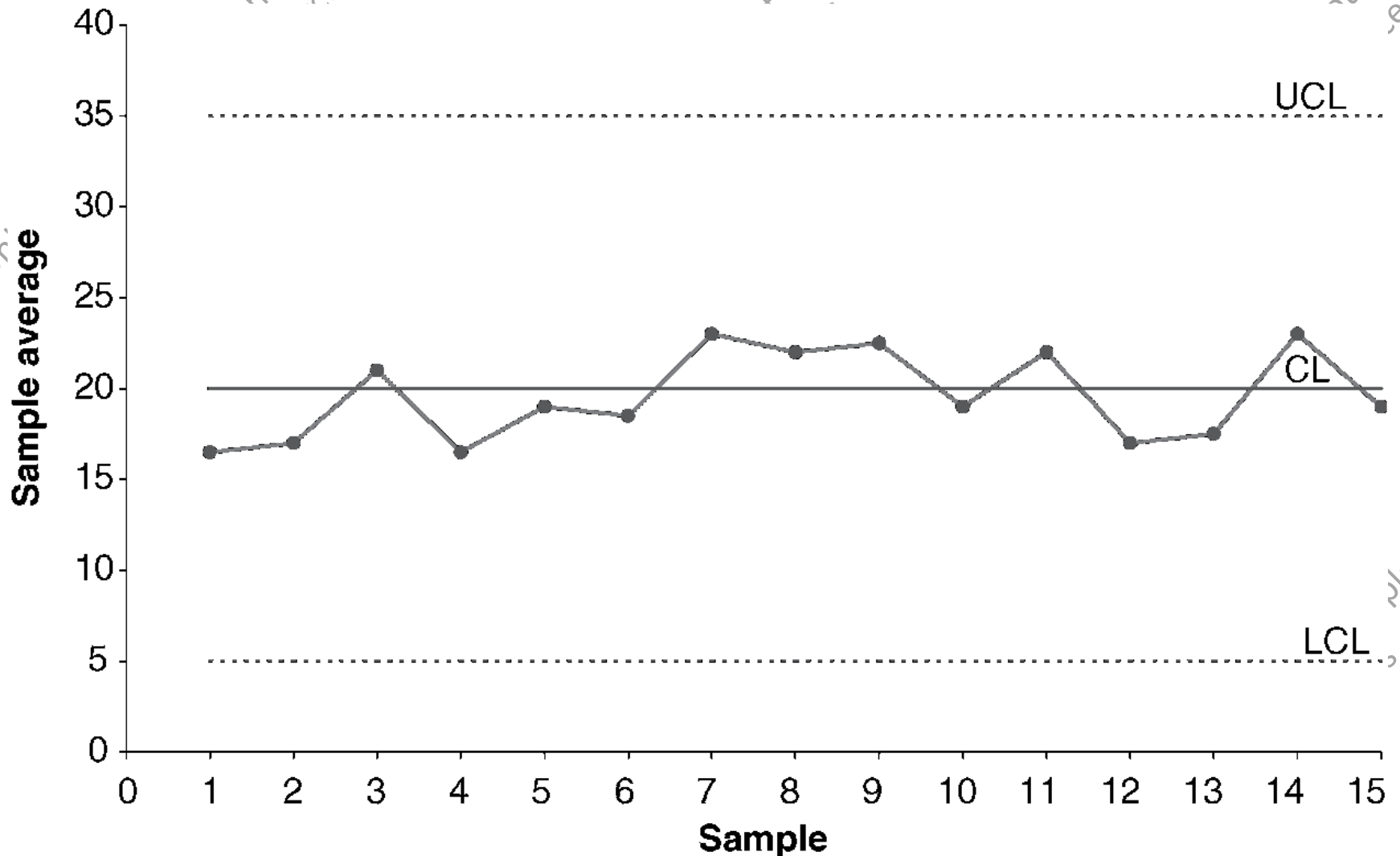
# Control Chart Patterns and Corrective

- Mixture Patterns (or the Effect of Two or More Populations)



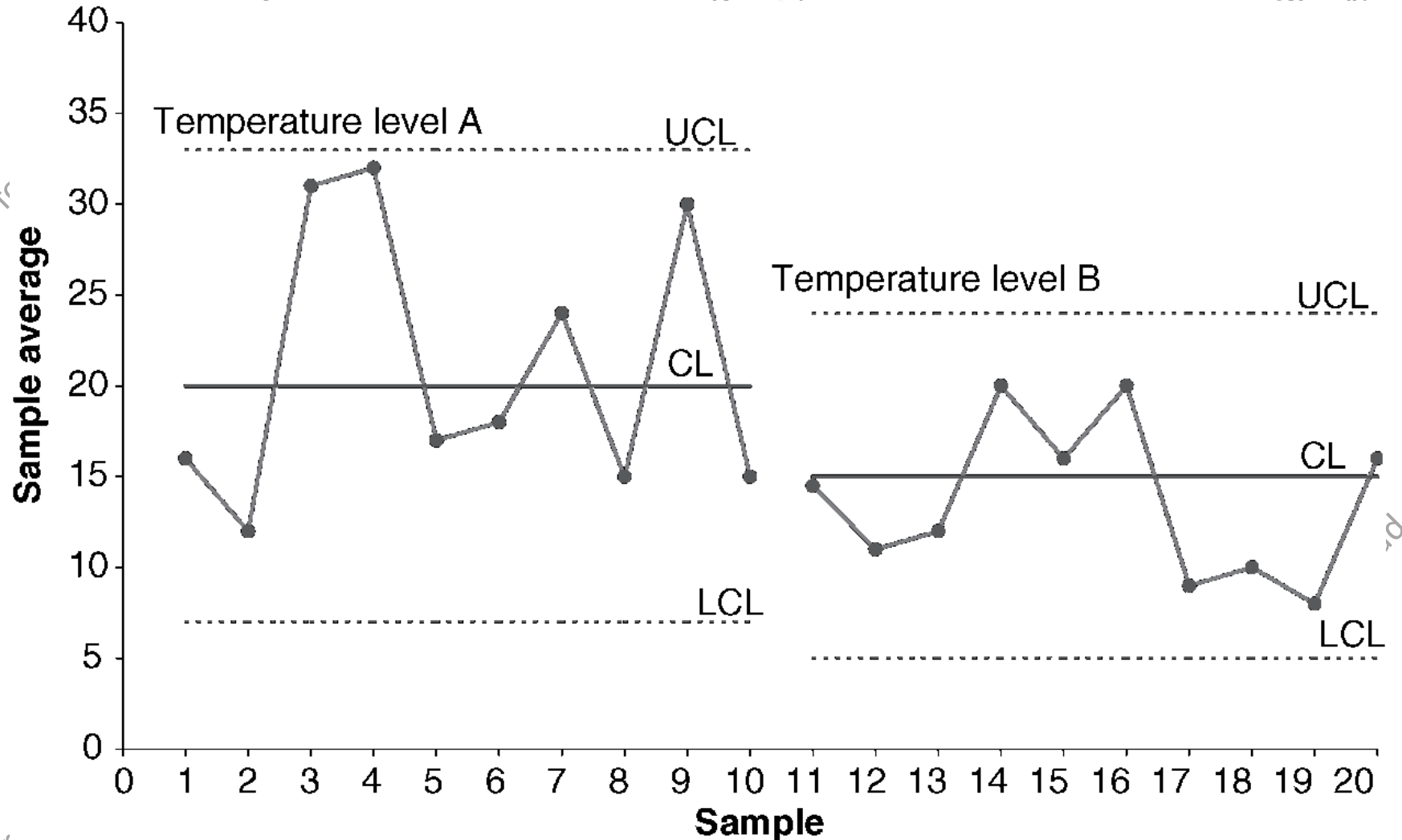
# Control Chart Patterns and Corrective

- Stratification Patterns



# Control Chart Patterns and Corrective

- Interaction Patterns



# Control Charts for the Mean and Range

- Development of the Charts
- Variable Sample Size
- Standardized Control Charts
- Control Limits for a Given Target or Standard
- Interpretation and Inferences from the Charts
- Control Chart Patterns and Corrective

# Development of the Charts

- **Step 1:** Using a pre-selected sampling scheme and sample size, record on the appropriate forms the measurements of the quality characteristic selected.
- **Step 2:** For each sample, calculate the sample mean and range.
- **Step 3:** Obtain and draw the centerline and the trial control limits for each chart.
- **Step 4:** Plot the values of the range on the control chart for range, with the centerline and the control limits drawn. Determine whether the points are in statistical control. If not, investigate the special causes associated with the out-of-control points and take appropriate remedial action to eliminate special causes.
- **Step 5:** Delete the out-of-control point(s) for which remedial actions have been taken to remove special causes and use the remaining samples to determine the revised centerline and control limits for the X- and R-charts.
- **Step 6:** Implement the control charts.

# Standardized Control Charts

The mean of the sample averages is found as

$$\bar{\bar{X}} = \frac{\sum_{i=1}^g n_i \bar{X}_i}{\sum_{i=1}^g n_i}$$

An estimate of the process standard deviation,

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^g (n_i - 1) s_i^2}{\sum_{i=1}^g (n_i - 1)}}$$

Now, for sample  $i$ , the standardized value for the mean,  $Z_i$ , is obtained from

$$Z_i = \frac{\bar{X}_i - \bar{\bar{X}}}{\hat{\sigma} / \sqrt{n_i}}$$

So, the standardized value for the range,  $k_i$ , is given by

$$k_i = \frac{r_i - d_2}{d_3} \quad \begin{array}{l} d_2 = \text{Relative Range} \\ d_3 = \sigma_w; r_i = \frac{R_i}{\hat{\sigma}} \end{array}$$

These values of  $k_i$  are plotted on a control chart with a centerline at 0 and upper and lower control limits at 3 and -3, respectively.

# Control Limits for a Given Target or Standard

The centerline and control limits based on these standard values for the  $\bar{X}$ -chart are given by  $\left(A = \frac{3}{\sqrt{n}}\right)$

$$CL_{\bar{X}} = \bar{X}_o$$

$$UCL_{\bar{X}} = \bar{X}_o + A\sigma_o$$

$$LCL_{\bar{X}} = \bar{X}_o - A\sigma_o$$

For the R-chart, the centerline is found as follows ( $\sigma_R = d_3\sigma_o$ )

$$CL_R = d_2\sigma_o = \bar{R}$$

$$UCL_R = D_2\sigma_o$$

$$LCL_R = D_1\sigma_o$$